

Exercise 3 – Case Study: Point of Sale System

Learning objectives: Understand how tests and quality objectives can be derived for a large scale system

Our goal today is the specification of a Point of Sale (POS) system for a gas station chain.

- Functions per Point of Sale
 - Listing the fuel prices of a petrol station
 - Changing fuel prices
 - Adding goods
 - ...
- Functions per petrol station chain / petrol station
 - Adding a gas station
 - Deleting a gas station



Task 1 – Test Objectives and Quality Attributes

You are responsible for the acceptance test of a POS system (e.g. for gas stations). It is the first time that you are introducing such a system. The system has been specifically developed for you.

1.1.) Define test objectives for your test project. Provide a justification your decision.

1.2.) Define the three most important quality attributes with the highest priority and justify your decision.

Quality Attribute 1	Quality Attribute 2	Quality Attribute 3

1.3.) What constructive/organizational quality assurance measures did you take in the run-up to the acceptance test to ensure smooth acceptance and high test object quality? List five examples and briefly explain why the measures led to higher quality!

1.4.) The following problems are observed during the acceptance test. Which of the problems represent failures, which represent defects? Why?

- a) The system reacts with delay when adding a new product
- b) The corporate identity color scheme is not applied consistently
- c) Displayed fuel prices are outdated

Task 2 – Errors

What are potential reasons for human made errors? Brainstorm as a team. List at least 5 reasons.

Task 3 – Traceability

What is traceability in testing? Why is maintenance of traceability important?
